Bhumika Sahu

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EDUCATION

Bhilai Institute of Technology **Durg, Chhattisgarh**
B. Tech (Computer Science and Engineering) *Graduation Date: Jul 2025*

SKILLS

GenAI & LLM Systems: LangChain, LangGraph, RAG Architecture, Vector Embeddings, Prompt Engineering, Token Optimization, Structured Output, Semantic Evaluation, RAGAS, Cross-Encoder Reranking, Working with Open-Source Models
Programming Languages: Python, Solidity, SQL
Backend Engineering: FastAPI, REST API Design, Server-Sent Events (SSE), Async Task Queues, Pydantic Validation, RBAC, Rate Limiting, Error Handling, Database Design (SQLite, PostgreSQL)
ML & Computer Vision: Scikit-learn, Keras, TensorFlow, OpenCV, MediaPipe, Sentence-Transformers, FAISS, BM25, Model Optimization
Web3 & Blockchain: Web3.py, Hardhat, Ethers.js, Smart Contracts, MetaMask, Infura, EVM Chains
DevOps & Infrastructure: Docker, Render, HuggingFace Spaces, Git/GitHub, GitHub Actions CI/CD, API Monitoring, Health Checks

WORK EXPERIENCE

VRIZE **RAIPUR, C.G**
Associate Software Engineer *Sept, 2025 – Present*

Meridian SDLC (AI-Driven Software Delivery Platform)

- Designed a FastAPI Server-Sent Events (SSE) streaming pipeline for incremental artifact generation, reducing perceived frontend latency from 45s to 2–3s while supporting 100+ concurrent generations.
- Built a durable asynchronous job persistence system using SQLite and task queues, ensuring reliable backlog streaming and artifact consistency.
- Enhanced GitHub integration by implementing repository/branch discovery and automated pull request workflows through 3 REST APIs, reducing manual setup by 65%.
- Developed context-aware requirement validation and deterministic LLM fallback mechanisms, reducing generation failures from 8% to <0.1% while improving output quality.
- Debugged and optimized critical FastAPI backend components involving concurrency, middleware, configuration isolation, and timeout handling, increasing automated test coverage from 68% to 94%.
- Implemented project catalog and Microsoft Teams transcription APIs with RBAC-enabled metadata persistence and comprehensive unit test coverage.

Training

- Completed hands-on GenAI bootcamp covering RAG systems, LangChain, LangGraph, FastAPI patterns, LLM integration, and prompt engineering best practices.

PROJECT EXPERIENCE

Personal Study Coach (GitHub) (Live-Link)
FastAPI · LangChain · LangGraph · FAISS · BM25 · Cross-Encoder · Streamlit · Docker · RAGAS
Production-Grade Hybrid-RAG Study Assistant with Autonomous Agent

- Architected 5-stage hybrid retrieval pipeline: LLM query rewriting → FAISS MMR (dense) → BM25Okapi (sparse) → Reciprocal Rank Fusion (k=60) → cross-encoder reranking (ms-marco-MiniLM-L-6-v2), achieving 92% answer relevance across evaluation benchmarks.
- Built autonomous LangGraph study-planner agent with 3 tool functions (search_docs, summarize_topic, generate_quiz_question), enabling multi-step reasoning and personalized study plan generation with Markdown export.
- Implemented RAGAS evaluation framework measuring faithfulness, answer relevancy, context precision, and context recall across 10 hand-written ground-truth Q&A pairs using Groq LLM as judge.
- Designed production backend serving 50+ concurrent users with per-session memory (20 messages), incremental FAISS persistence, path-traversal prevention, 50 MB upload cap, and JSON corpus storage (no pickle).
- Wrote 37 automated tests (11 retriever, 16 API integration, 10 agent tool tests) with state-injection fixtures and mocked LLM dependencies.
- Deployed as containerized two-service stack (Docker Compose) with FAISS data in named volumes; live on HuggingFace Spaces.

Redrob Candidate Ranker — Hackathon (GitHub)
Python · Sentence-Transformers · NumPy · Cosine Similarity · Heap-Based Ranking
Rank 100K Candidates Against a JD in <5 Min on CPU — No LLM Calls | github.com/Bhumika987/redrob-candidate-ranker-Hackathon-

- Engineered deterministic candidate-ranking pipeline that scores 100K profiles and outputs top-100 CSV in under 5 minutes on CPU, using precomputed all-MiniLM-L6-v2 embeddings (384-dim, float32) and a fixed-size heap for memory-efficient streaming.
- Designed multi-factor scoring formula: title/career fit (0.35), semantic cosine similarity (0.25), experience curve (0.15), production-evidence keywords (0.15), and location fit (0.10) — with behavioural multipliers and consistency checks (honeypot detection, impossible-profile flagging).
- Implemented soft/hard disqualifiers for services-only careers, title-chasing patterns, and LLM-wrapper-only profiles, with multiplicative penalty stacking and recency-weighted career history scoring.
- Built negative-signal detection in semantic scoring (penalizes aspirational vs. shipped-product language) and 4-tier consistency verification (expert-skill-with-zero-duration honeypots, career-duration vs. YoE mismatch, impossible date ranges).

MultiChain Gas Tracker API (GitHub) (Live-Link)
FastAPI · Web3.py · SQLite · Docker · Render · GitHub Actions CI/CD
Real-Time EVM Gas Fee Intelligence Platform — Live in Production | multi-chain-gas-tracker-api.onrender.com

- Restructured monolithic FastAPI codebase into modular production architecture (routers, services, repositories, schemas) with app-factory pattern and Pydantic models across 13 REST endpoints.
- Built RPC provider resilience layer with connection pooling, exponential backoff retry (3 retries), multi-provider failover, and environment-driven secrets management — achieving 99.2% uptime across 500+ daily API calls.
- Implemented production observability: health/readiness probes, endpoint-specific rate limiting (60 RPM), request-ID middleware for end-to-end tracing, masked RPC URL logging, and Prometheus-compatible /metrics endpoint.
- Added background historical base-fee collector persisting 30-day gas trends to SQLite with WAL mode, indexes, and bounded query optimization for time-series analytics across 7 EVM chains.
- Established pytest suite with 19 tests covering health endpoints, chain config, RPC masking, gas service logic, and failure handling; deployed via GitHub Actions CI/CD and Docker on Render.

Real-Time ASL Recognition System (GitHub)
TensorFlow · Keras · OpenCV · MediaPipe · CNN

- Trained CNN for real-time American Sign Language recognition across 32+ gesture classes (A–Z + common phrases) using custom-collected dataset and MediaPipe hand-tracking preprocessing.
- Engineered end-to-end inference pipeline achieving <200ms latency: webcam capture → MediaPipe hand isolation → CNN classification → real-time gesture translation overlay.

CERTIFICATIONS

- Ultimate RAG Bootcamp Using LangChain, LangGraph & Langsmith (Certificate)
- Shell Script for Beginners (Certificate)
- Learning Linux Basics Course & Labs (Certificate)
- Web Scraping in Python (Certificate)